

STATEMENT OF WORK LOW BACKGROUND INFRARED (LBIR) SERVICE SUPPORT

1. MISSION BACKGROUND

The Contractor shall provide essential research, standards development and calibration service assistance utilizing Government-furnished equipment at the National Institute of Standards and Technology (NIST) Low Background Infrared (LBIR) facility. This support ensures the accuracy, repeatability, and SI-traceability of Kill Vehicle (KV) sensor-seekers utilized in Missile Defense Agency (MDA) interceptor systems

1.1. Operational Context

Ground-based Midcourse Defense (GMD) and Aegis SM-3 interceptor systems rely on Kill Vehicle (KV) sensor-seekers to discriminate and intercept lethal objects launched via intermediate-range and intercontinental ballistic missiles. Effective target discrimination and engagement require onboard KV sensors to precisely measure the radiometric characteristics of potential threats.

1.2. KV Testing and Evaluation

To ensure mission success, KV sensors undergo rigorous characterization and mission scenario testing. Laboratories utilize blackbodies and infrared (IR) emitter arrays to simulate the complex, real-time IR signatures of reentry vehicles, debris, and countermeasures. This stimulus enables the comprehensive evaluation of KV guidance algorithms and discrimination capabilities.

The primary test facilities supported by this calibration effort include:

Facility / Organization	Test Chamber Designation	Purpose
Arnold Engineering Development Complex (AEDC)	AEDC Test Chambers	KV mission scenario testing
Johns Hopkins University Applied Physics Laboratory (JHU-APL)	Infrared Seeker Space Calibration and Test (ISSCAT)	KV mission scenario testing
Next Generation Interceptor (NGI)	NGI Test Chambers	KV mission scenario testing
RTX (Raytheon)	Low-background Scanning Point Source (LSPS)	KV sensor production characterization

1.3. Calibration and Traceability Requirements

To guarantee that the aforementioned test chambers provide consistently repeatable and accurate measurement data across production cycles, periodic calibration against established measurement standards is mandatory. All calibration standards must maintain strict traceability to the System International (SI) units of radiometric power and wavelength, as governed by NIST.

1.4. The LBIR Facility

The Air Force Metrology and Calibration (AFMETCAL) directorate established the LBIR facility at NIST and serves as the Department of Defense (DoD) MDA executing agent. Located at the NIST Sensor Science Division in Gaithersburg, MD, the LBIR facility provides the IR user community with a foundational calibration base, executing both spectral and broadband measurements within a 20-kelvin background environment. Broadband measurements are specifically utilized for the calibration of blackbodies in radiance mode.

1.5. Contractor Execution Requirements

The Contractor is responsible for the technical operation, maintenance, and data synthesis of the LBIR facility's specialized hardware. Performance requires on-site expertise in cryogenic radiometry, adherence to NIST safety protocols, and rigorous data management to maintain SI-traceability.

1.5.1. Technical Operations and Measurement

The Contractor shall provide the subject matter expertise required to execute complex radiometric measurements. This includes, but is not limited to:

Requirement Area	Specific Tasking
Radiometer Operation	Setup, alignment, and operation of the BXR, MDXR, and UXR cryogenic transfer radiometers for both on-site and field-deployed calibrations.
Environmental Control	Monitoring and maintaining the 20-kelvin background environment during measurement cycles to ensure thermal stability and data integrity.
Measurement Execution	Performing spectral and broadband calibrations on user detectors, optical components, and blackbody sources in accordance with established NIST protocols.
Data Processing	Utilizing Government-furnished software to reduce raw measurement data, calculate uncertainties, and generate draft calibration reports.

1.5.2. Laboratory Maintenance and Stewardship

The Contractor shall ensure the operational readiness of the LBIR laboratory by performing the following:

Activity	Description
Equipment Maintenance	Performing routine preventative maintenance on cryogenic systems, vacuum pumps, and specialized optical hardware in Building 221.
Inventory Management	Maintaining an accurate inventory of Government-furnished equipment (GFE) and identifying requirements for replacement parts or consumables.
Safety Compliance	Adhering to all NIST and DoW safety regulations, specifically those governing cryogenic liquids, high-vacuum systems, and laser operations.

1.5.3. Site Access and Personnel Constraints

Given the location and nature of the mission, the following execution constraints apply:

- **Work Location:** All primary tasks must be performed on-site at the National Institute of Standards and Technology (NIST) campus, 100 Bureau Drive, Gaithersburg, Maryland, Building 221.
- **Access Protocols:** Contractor personnel must satisfy all NIST-specific security requirements for facility access and utilize Government-furnished Information Technology (IT) systems in accordance with IL5 security protocols.
- **CUI Handling:** The Contractor shall handle all measurement data and technical reports as Controlled Unclassified Information (CUI) unless otherwise directed by the Contracting Officer's Representative (COR).

2. SCOPE OF WORK

The Contractor shall provide comprehensive technical support for the calibration, operation, maintenance, and modernization of the Low Background Infrared (LBIR) Facility. The LBIR serves as the national reference standard for infrared radiation measurements in cryogenic environments. The Contractor's efforts are divided into three primary activities: Routine Operation and Maintenance, Calibration Support, and Metrology Infrastructure Modernization.

2.1. Routine Operation and Maintenance

This activity maintains the continuous operational readiness and day-to-day functionality of essential systems at the LBIR Calibration Facility. This ensures the reliable delivery of precise and reproducible calibration data critical to Missile Defense Agency (MDA) test chamber assessments.

Supported System	Description
Missile Defense Transfer Radiometer (MDXR) & User Transfer Standard Radiometer (UXR)	Portable, precision state-of-the-art transfer-standard radiometers engineered to deliver accurate on-site calibration of customer test facilities, ensuring consistent measurement reliability across diverse operational environments.
10 Centimeter Collimator (10CC) & 30 Centimeter Collimator (30CC)	Precision optical systems that facilitate the transfer of calibrations from the LBIR's Absolute Cryogenic Radiometer (ACR) to the MDXR and UXR instruments, ensuring accurate traceability.
Broadband Calibration Chamber (BCC)	Specialized advanced calibration facility that calibrates vacuum cryogenic blackbody sources against the ACR reference standard.
Spectral Calibration Chamber (SCC)	Specialized facility engineered for the spectral calibration of infrared detectors under ultra-low background conditions.

2.2. Calibration Support

This activity provides scheduled on-site and off-site calibration services to validate measurement uncertainties for both test chambers and blackbody sources, ensuring precise characterization and reliable performance of Standard Missile-3 (SM-3) sensor systems. Services leverage national

reference standards, highly qualified technical personnel, and complete traceability to the International System of Units (SI).

Category	Description
Cryogenic Test Chambers	Low Background Scanning Point Source (LSPS) 11a, LSPS 16a, and the Infrared Seeker Space Calibration and Test (ISSCAT) facilities.
Infrared Sources	Blackbody sources and related infrared emission systems, offering stable, traceable reference signals across the infrared spectrum.
Detection Systems	Secondary standard radiometers, pyroelectric radiometers, and infrared detector assemblies.

2.3. Metrology Infrastructure Modernization

This initiative strengthens calibration capabilities and operational efficiency for Kill Vehicle (KV) sensor testing while reducing production disruptions.

Objective Area	Description
System Upgrades	Upgrade LBIR systems, implement modern calibration frameworks, and refine tools and procedures to reduce chamber and blackbody calibration uncertainty.
Measurement Standards	Design and produce advanced measurement standards that meet current and future KV mission needs.
Technical Support	Provide specialized scientific and technical metrology support to MDA stakeholders.
Innovation & Cross-Validation	Develop innovative measurement methods, technologies, and analytical approaches. Conduct cross-validation and calibration of laboratory reference standards.

3. TECHNICAL REQUIREMENTS AND MAINTENANCE TASKS

The Contractor shall execute comprehensive operations, rigorous preventative maintenance, and continuous improvement initiatives for the LBIR facility's highly specialized cryogenic and radiometric systems. Additionally, the Contractor shall manage the authorized procurement of necessary materials and provide detailed monthly reporting to ensure uninterrupted mission support.

3.1. Operations and Maintenance (O&M) of LBIR Infrastructure

To fulfill the requirements of Section 2.1 (Routine Operation and Maintenance), the Contractor shall perform specific, scheduled maintenance on facility hardware. Close contact shall be maintained with the physical plant to ensure all support services (e.g., chilled water, electrical power) function during operations.

The Contractor shall perform the following subsystem-specific maintenance tasks in accordance with the applicable documented standard operating procedures (SOPs).

Table 3.1: Refrigeration Systems Maintenance

Subsystem	Maintenance Action	Frequency/Requirement
Linde Inc. RS & RSX Compressors	Evacuate and replace charcoal filters	Every 10,000 hours of operation
	Replace rotary scroll compressor.	Every 100,000 hours of operation
	Check for operational failure and oil blow-through	Periodically during operation
	Execute start-up and shut-down sequences	Per documented Standard Operating Procedures (SOPs)
Linde Inc. 1620 Cold Boxes	Re-lubricate all essential parts	Every 250 hours of operation
	Bake out charcoal filters	Every 3,000 hours of operation
	Rebuild piston and valve systems	Every 3,000 hours of operation
	Monitor and check for He leaks	Periodically to minimize He loss
	Execute start-up and shut-down sequences	Per documented SOPs

Table 3.2: Chamber and Pumping Systems Maintenance

System	Contractor Responsibilities
Vacuum Integrity	Maintain vacuum parameters through strict cleanliness protocols, leak checking hardware, and repairing or replacing vacuum hardware as necessary.
Electrical Systems	Maintain and improve electrical cabling used for the control of test hardware and the monitoring of internal chamber temperatures.
Cryogenic Systems	Maintain cryogenic systems utilizing leak detection systems and routine troubleshooting procedures. Improve systems through the modeling, design, and construction of He cryostats and heat exchangers.
Pumping Systems	Maintain all rotary vane, oil-free rotary scroll, turbo, and cryo-pumps strictly according to the instructions provided in their respective manufacturer manuals.

4. PROCUREMENT REQUIREMENTS

The Contractor shall develop and implement procedures to purchase materials and services required to support the activities outlined in Sections 2.1 through 2.3.

Requirement Area	Contractor Responsibilities
Procurement Authority	The Contractor is authorized to procure all required services, equipment, and any other items required by the Government to support the technical tasks.
Cost Reimbursement	Each Project Plan will specify the authorized procurement amount required by the Government for which the Contractor will be reimbursed.
Procurement Reporting	All purchases shall be reported on a monthly basis (see Section 5. Deliverables).

5. PLACE OF PERFORMANCE AND RESOURCES

The Contractor shall primarily provide services on-site at the National Institute of Standards and Technology (NIST) campus.

5.1. Primary Performance Site

Location Details	Description
Primary Address	NIST LBIR Facility, 100 Bureau Drive, Gaithersburg, Maryland, Building 221.
Off-Site Permission	Any or all tasks may be performed off-site only with the explicit permission of the Contracting Officer's Representative (COR).

5.2. Government Furnished Property (GFP) and Office Space

All materials provided by the Government during the performance of this contract remain the property of the Government and shall be surrendered to the Government upon completion or termination of this requirement.

Resource Category	Items Provided by NIST
Office Space & Comms	Office space, telephones (local/long-distance), faxes, and copiers.
IT Equipment	Personal computers, internet access, and laboratory data access.
Software & Services	Ordinary business and special application software, and reproduction services.

5.3. Travel and Off-Site Calibration Locations

5.3.1. Travel Authorization

The Contractor may be required to travel in support of calibration and validation activities conducted by LBIR. All travel shall receive prior written approval from the Executing Agent for the Missile Defense Agency.

5.3.2. Reimbursable Expenses

The Government shall reimburse the Contractor for all necessary and authorized travel expenses incurred pursuant to this Contract. Reimbursable costs for lodging, meals, and incidental expenses must be reasonable and shall not exceed the applicable daily maximum per diem rates established in the Federal Travel Regulations (FTR) issued by the General Services Administration.

5.3.3. Subsistence

Subsistence is authorized in accordance with the Federal Travel Regulations.

5.3.4. Documentation Requirements

All invoices submitted for travel reimbursement shall include itemized, paid receipts for each expense claimed.

5.3.5. Travel Locations

The contractor shall be available to assist in off-site calibrations at the following locations and any other location where MDA and Air Force customers may be located:

Customer/Facility	Location
RTX	Tucson, AZ
JHU-APL	Laurel, MD
AEDC	Tullahoma, TN
NGI	Huntsville, AL

6. DELIVERABLES SCHEDULE

The following deliverables summarize the expected outputs across all operational and modernization activities.

Deliverable	Associated Task	Description
Operational Availability	Routine Operations	Guaranteed operational availability of MDXR, UXR, 10CC, 30CC, BCC, and SCC systems to support ongoing precision calibration and testing.
Calibration Data & Reports	Calibration Support	NIST-traceable calibration services, comprehensive measurement data acquisition, thorough data analysis, and complete calibration documentation including uncertainty evaluations.
Advanced Systems & Standards	Modernization	Development of next-generation calibration equipment and precision standards. Procured commercial components and supporting hardware infrastructure.
Technical Documentation	Modernization	Detailed procedures, methodologies, and uncertainty analyses. Custom software and automated measurement tools.
Monthly Purchase Report	Procurement	A report detailing all acquired items/services. Must include: description, manufacturer, part number, cost, and a justification describing how the purchase supports the SOW tasks.

7. PERSONNEL REQUIREMENTS

7.1. Experience

Effective support of these facilities will require a team of personnel with experience in the following areas:

- 7.1.1. Large cryogenic refrigerator systems.
- 7.1.2. Storage and handling of cryogenic liquids such as nitrogen and helium.
- 7.1.3. Design, maintenance and use of large scale, high vacuum chambers and associated equipment.
- 7.1.4. Design, maintenance and use of thermal infrared optical equipment, such as cryogenic radiometers, electro-optic detectors, cryogenic mirrors and opto-mechanical equipment.

- 7.1.5. Maintenance and use of visible and infrared laser systems.
- 7.1.6. Maintenance and use of cryogenic (20 Kelvin) vacuum Fourier transform infrared radiometers.
- 7.1.7. Hardware and software design and implementation for effective data acquisition and automation.
- 7.1.8. Data reduction, error analysis and report writing.
- 7.1.9. Design (using Zemax optical design software and Solid Works mechanical design software), assembly, and operation of cryogenic imaging, spectroscopic, and radiometric instrumentation and associated test and calibration procedures.
- 7.1.10. Overall good mechanical skills for the routine maintenance of vacuum pumps, clean room facilities, and other simple hardware that needs routine maintenance.
- 7.1.11. Overall good computer operating systems, hardware, security and networking skills for compliance with NIST internet technology security requirements.

7.2. Education and Experience Requirements:

The contractor shall have as a minimum the following key personnel:

- 7.2.1. Two highly experienced specialized senior principal scientist or senior engineer with a Ph.D. degree, from an accredited college or university, in a physical science, mathematical or engineering field with at least 10 years of experience in building high vacuum cryogenic systems for infrared instrumentation and making measurements of infrared irradiance and radiance using cryogenic Fourier transform spectrometers, cryogenic filter radiometers, cryogenic blackbodies, and cryogenic electrical substitution radiometers. One of these must be able to provide 1.0 level of effort (LOE) per year, and the other at least 0.5 LOE per year.
- 7.2.2. One Senior MA/MS-level Scientist with an MA or MS degree in a physical science, mathematics, or engineering discipline, and at least ten years of experience developing systems for infrared instrumentation and performing infrared irradiance and radiance measurements using Fourier transform spectrometers.
- 7.2.3. One Senior Technician with a strong technical background and a minimum of five years' experience working with cryogenic and high-vacuum equipment, maintaining Linde helium gas refrigerators, and constructing electronic equipment according to provided wiring diagrams with professional-grade quality and workmanship.

7.3. NIST Domestic Guest Researcher Agreement

All contractor personnel must have an approved Domestic Guest Research Agreement (NIST Form 1296) prior to performing any work in support of this contract. NIST Domestic Guest Researcher Agreement is required to collaborate with NIST on research and development projects of this contract.

8. SECURITY REQUIREMENTS

The Contractor shall comply with all applicable NIST, Department of Commerce, and Department of Defense (DoD) security regulations required for performance at government facilities and designated customer sites.

Security Category	Contractor Responsibilities
Personnel Security	Contractor personnel must successfully complete the required background investigations (e.g., Tier 1 / NACI) to obtain a NIST Personal Identity Verification (PIV) card and unescorted campus access. Additional security clearances may be required for specific off-site MDA or Air Force facilities.
Information Security (CUI)	The Contractor is authorized to process and generate Controlled Unclassified Information (CUI). All CUI must be handled, stored, and transmitted in accordance with DoD and NIST security protocols (including NIST SP 800-171 standards where applicable).
Physical Security	Personnel shall visibly display issued identification badges at all times while on Government property and adhere to all facility access control procedures.
IT System Access	Access to NIST and DoW/DoD laboratory data networks requires completion of mandatory annual Information Assurance (IA) and Cyber Awareness training.
Export Control Compliance	Due to the nature of missile defense sensor technology, the Contractor shall strictly comply with all International Traffic in Arms Regulations (ITAR) and Export Administration Regulations (EAR) when handling technical data.

9. COLLABORATIVE PARTNERSHIP

The Contractor shall work in collaborative partnership with the NIST Sensor Science Division, Remote Sensing Group. Both parties will maintain an ongoing dialogue to coordinate day-to-day activities and establish the technical priorities required to address the Major Tasks and specific objectives identified in each Air Force Project Plan. This collaborative approach ensures that the Contractor's expertise is fully integrated with NIST mission goals and strategic planning.

10. MEETINGS AND REPORTING REQUIREMENTS

All required documentation can be in contractor's format. Draft copies of required technical documentation/reports identified in this SOW shall be provided at least one week prior to any scheduled reviews/acceptance.

- 10.1. Reporting: A monthly Activities Summary Report shall be prepared that provides a summary of activities, technical progress and anticipated completion timeframes on each Major Task identified in paragraph 2 of this Statement of Work and each specific task in each Project Plans.
- 10.2. The contractor personnel shall provide a weekly oral report of progress to the COR. The oral report shall include the following for each week:

- 10.2.1. Activities accomplished; any problems encountered; summary of data; issues; risks; and recommendations that were brought to the attention of the COR.
- 10.2.2. Planned activities for the next activity period.
- 10.3. Formal Meetings/Reviews: The contractor shall participate in progress meetings/reviews conducted at NIST with the Executing Agent to review all tasking progress and schedule.
- 10.4. Informal Meetings/Reviews: The contractor shall participate in any informal technical interchange meetings deemed necessary by the Infrared Technology Group Leader.

11. PROJECT PLANS

The contractor shall receive specific direction/tasking from Contracting Officer (KO) or designated Contracting Officer's Representative (COR) using Project Plans. Each Project Plan provides the contractor specific direction, a performance standard, and a government approved monitoring method.